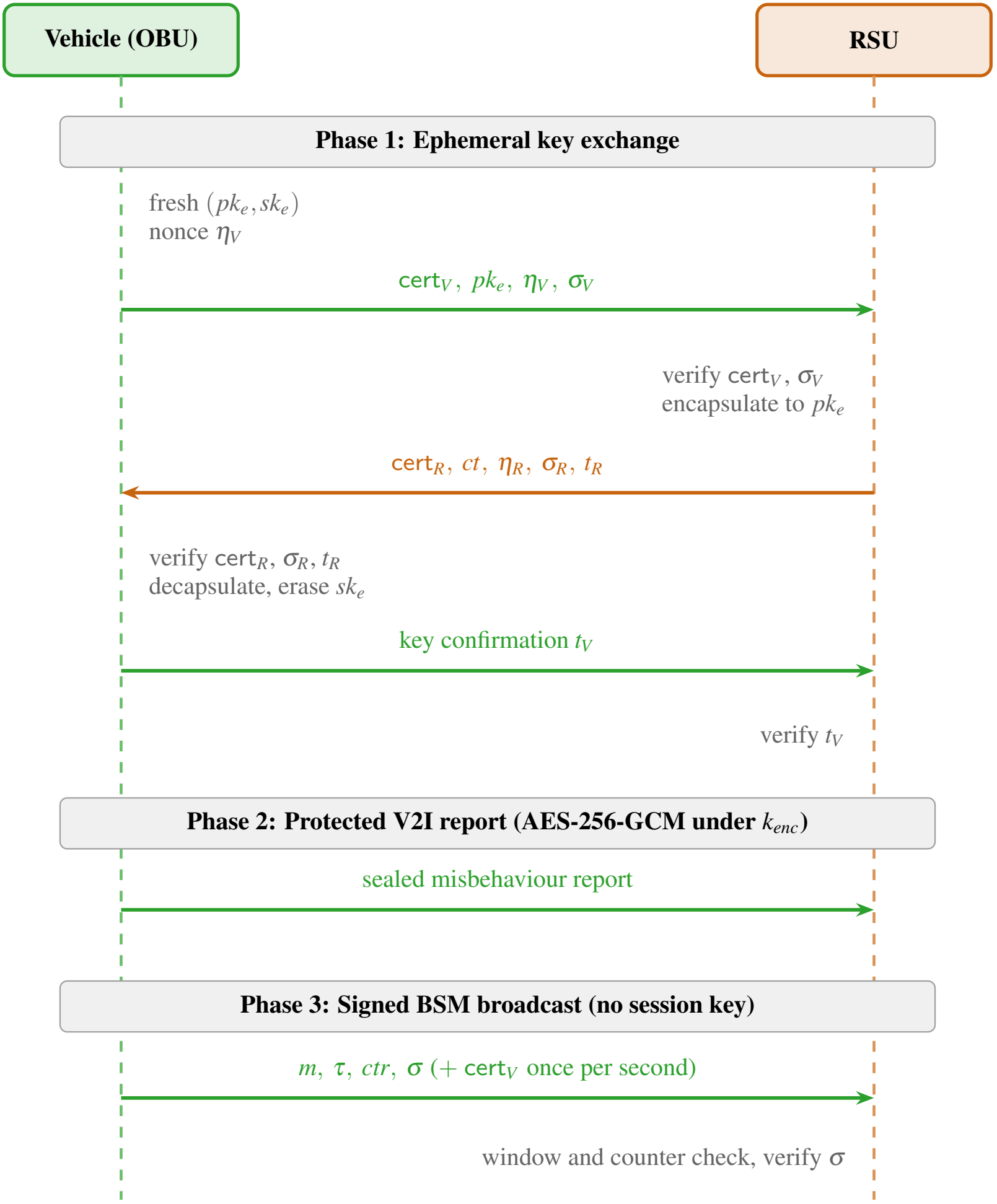

Vehicle (OBU)
RSU
Phase 1: Ephemeral key exchange
fresh (pk_e, sk_e)
nonce η_V
cert_V, pk_e, η_V, σ_V
verify cert_V, σ_V
encapsulate to pk_e
cert_R, ct, η_R, σ_R, t_R
verify cert_R, σ_R, t_R
decapsulate, erase sk_e
key confirmation t_V
verify t_V
Phase 2: Protected V2I report (AES-256-GCM under k_{enc})
sealed misbehaviour report
Phase 3: Signed BSM broadcast (no session key)
m, τ, ctr, σ (+ cert_V once per second)
window and counter check, verify σ